

ASSIGNMENT NO.7: Subqueries and Nested Queries

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: Course and StudentResult.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

StudentResult Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	FOREIGN KEY
City	VARCHAR(30)	
Marks	INT	

Insert at least 4 records into Course and 10 records into StudentResult.

Query-

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR (50) NOT NULL,  
    Fees DECIMAL(10,2) NOT NULL  
);
```

```
CREATE TABLE StudentResult (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    CourseID INT,  
    City VARCHAR(30),  
    Marks INT,  
    FOREIGN KEY (CourseID) REFERENCES Course (CourseID)  
);
```

Q2) Write a MySQL query to display students whose marks are greater than the average marks of all students.

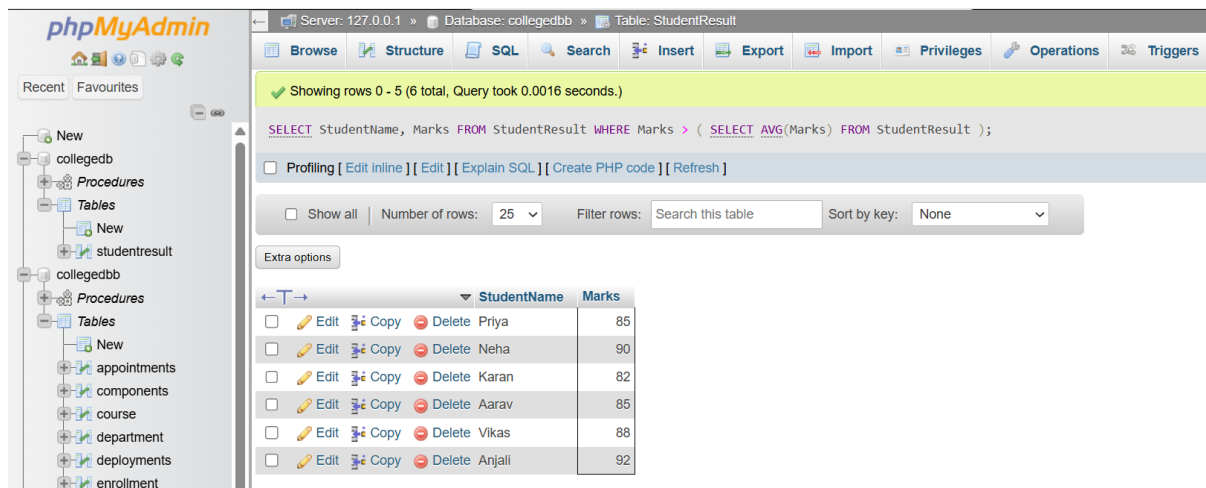
The output should contain:

Column
StudentName
Marks

Query-

```
SELECT StudentName, Marks
FROM StudentResult
WHERE Marks > (
    SELECT AVG(Marks)
    FROM StudentResult
);
```

Output-



Showing rows 0 - 5 (6 total, Query took 0.0016 seconds.)

```
SELECT StudentName, Marks FROM StudentResult WHERE Marks > ( SELECT AVG(Marks) FROM StudentResult );
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	StudentName	Marks
☐	Priya	85
☐	Neha	90
☐	Karan	82
☐	Aarav	85
☐	Vikas	88
☐	Anjali	92

Q3) Write a MySQL query to display all students enrolled in the BCA course using a subquery.

The output should contain:

Column
StudentID
StudentName
CourseID
Marks

Query-

```
SELECT StudentID, StudentName, CourseID, Marks
FROM StudentResult
WHERE CourseID = (
    SELECT CourseID
    FROM Course
    WHERE CourseName = 'BCA'
);
```

Output-

The screenshot shows the phpMyAdmin interface with the 'StudentResult' table selected. The query box contains the following SQL query:

```
SELECT StudentID, StudentName, CourseID, Marks FROM StudentResult WHERE CourseID = ( SELECT CourseID FROM Course WHERE CourseName = 'BCA' );
```

The result shows 1 row:

StudentID	StudentName	CourseID	Marks
6	Karan	106	82

Q4) Write a MySQL query to display each student with the overall average marks using a subquery in the SELECT clause.

The output should contain:

Column
StudentName
Marks
AverageMarks

Query-

SELECT

StudentName,

Marks,

(SELECT AVG(Marks) FROM StudentResult) AS AverageMarks

FROM StudentResult;

Output-

The screenshot shows the phpMyAdmin interface with the 'StudentResult' table selected. The query box contains the following SQL query:

```
SELECT StudentName, Marks, (SELECT AVG(Marks) FROM StudentResult) AS AverageMarks FROM StudentResult;
```

The result shows 10 rows:

StudentName	Marks	AverageMarks
Rahul	75	80.5000
Priya	85	80.5000
Amit	60	80.5000
Neha	90	80.5000
Riya	70	80.5000
Karan	82	80.5000
Aarav	85	80.5000
Sneha	78	80.5000
Vikas	88	80.5000
Anjali	92	80.5000

Q5) Write a MySQL query using a correlated subquery to display students who scored more than their own course average.

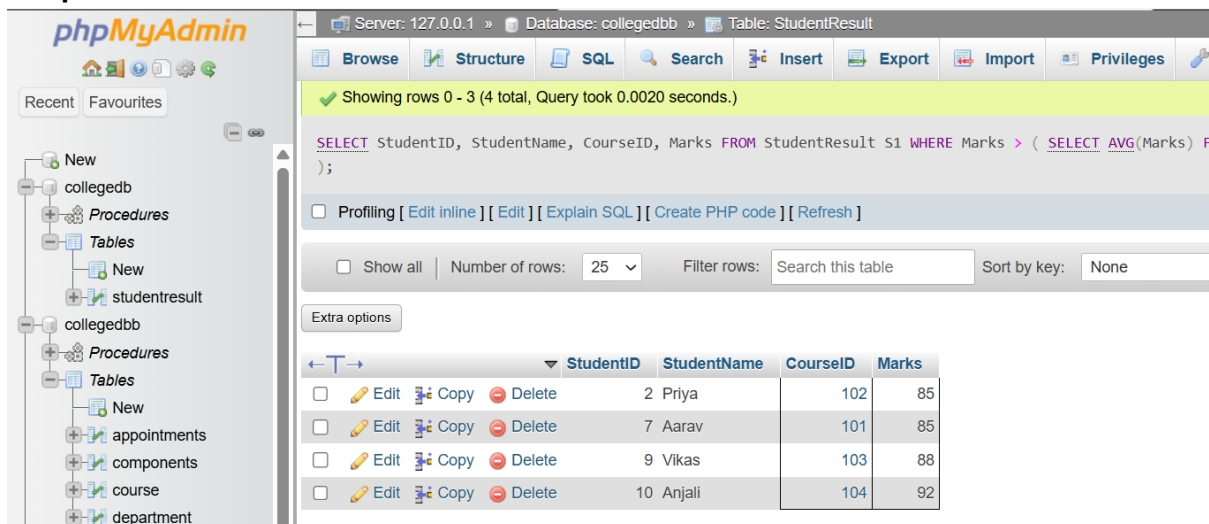
The output should contain:

Column
StudentID
StudentName
CourseID
Marks

Query-

```
SELECT StudentID, StudentName, CourseID, Marks
FROM StudentResult S1
WHERE Marks > (
    SELECT AVG(Marks)
    FROM StudentResult S2
    WHERE S2.CourseID = S1.CourseID
);
```

Output-



StudentID	StudentName	CourseID	Marks
2	Priya	102	85
7	Aarav	101	85
9	Vikas	103	88
10	Anjali	104	92

Q6) Write MySQL queries to display courses having students and courses having no students.

Use:

Operator
EXISTS
NOT EXISTS

Query-

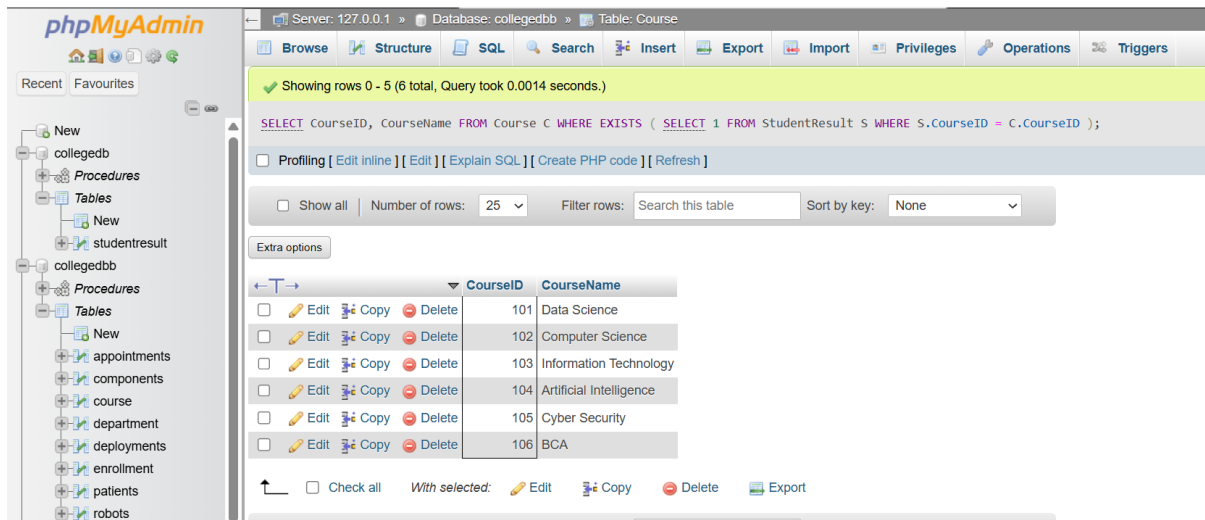
```
SELECT CourseID, CourseName
FROM Course C
WHERE EXISTS (
    SELECT 1
    FROM StudentResult S
    WHERE S.CourseID = C.CourseID
);
```

```

SELECT CourseID, CourseName
FROM Course C
WHERE NOT EXISTS (
    SELECT 1
    FROM StudentResult S
    WHERE S.CourseID = C.CourseID
);

```

Output-



Server: 127.0.0.1 » Database: collegedb » Table: Course

Showing rows 0 - 5 (6 total, Query took 0.0014 seconds.)

```

SELECT CourseID, CourseName FROM Course C WHERE EXISTS ( SELECT 1 FROM StudentResult S WHERE S.CourseID = C.CourseID );

```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	CourseID	CourseName
☐ Edit Copy Delete	101	Data Science
☐ Edit Copy Delete	102	Computer Science
☐ Edit Copy Delete	103	Information Technology
☐ Edit Copy Delete	104	Artificial Intelligence
☐ Edit Copy Delete	105	Cyber Security
☐ Edit Copy Delete	106	BCA

Check all | With selected: Edit Copy Delete Export